

SHOVAN SHAKYA

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Education

University of South Florida

Bachelors of Science in Electrical Engineering

Tampa, Florida

May 2025

Skills

Languages: C, C++, Python, Verilog

Software Development Tools: Zephyr RTOS, OpenCV, CUDA, Vivado, ROS2, CMake, Git

CAD Tools: Altium, KiCAD

Hardware: Oscilloscope, Logic Analyzer

Experience

REEKON Tools

Boston, Massachusetts

Embedded Software Engineer

June 2025 - Current

- Serving as the primary firmware developer for the T1R Range, contributed to Kickstarter launch (above \$300K preorders), building upon the existing Zephyr RTOS digital tape firmware to integrate LDM and IMU functionality.
- Implemented CUDA accelerated Goertzel filtering across pixels to isolate LED blink frequencies in video streams, increasing object-tracking throughput from 5 Hz (CPU) to 40 Hz (GPU).
- Developed UART driver for Bosch LDM, I2C drivers for GPIO Expander and Battery Charger.
- Improved display driver refresh rate from 8 Hz to 30 Hz and utilized semaphores for tear free LCD update.
- Utilized stepper motor and encoder system to build LDM accuracy test platform.
- Implemented dual-axis leveling from accelerometer data with 30 Hz update rate and LPF for stable visualization.
- Built a C++ camera-laser triangulation pipeline with LUT based multi angle calibration to estimate plane orientation, using named pipes for interprocess communication with a Python controlled galvo laser.
- Developed Kalman Filter based distance estimation using IMU to auto annotate measurements.

Jaycon Systems

Palm Bay, Florida

Firmware Engineering Intern

September 2024 - May 2025

- Enhanced STM32 firmware in C by reducing SPI EMI for FCC compliance, adding ADC based temperature logging at 2 Hz, and integrating SPI double buffering to prevent DMA buffer overwrites.
- Exposed ADC (thermistor, VIN) and GPIO data to desktop app via C++ and SWIG-generated C# bindings.
- Optimized scheduler parameters, increasing RFID read rates (10 Hz) preserving high performance LED control (45 Hz).
- Architected a Zynq SoC image processing state estimation solution for a NASA SBIR application.
- Developed and tested a connected component labeling (CCL) algorithm in C for ARM Cortex-A9 processor to extract target blob coordinates at a 15 Hz update rate from 720p camera.
- Using High Level Synthesis (C++) to implement FPGA accelerated realtime image processing on Zynq 7000 SoC.
- Developing a CCL algorithm in SystemVerilog and integrated dual-port BRAM to transfer pixel coordinate data from custom IP to the Zynq processor in C++.

Universal Creative

Orlando, Florida

Sensor Fusion Intern

May 2024 - August 2024

- Developed STM32 firmware in C for IMU interfacing over I2C, communicating with other STM32 modules via CANBUS, and transfer IMU data to a computer using UDP via Ethernet SPI.
- Designed a 4-layer STM32-based PCB (2cm x 2cm) with BNO088 IMU, Ethernet SPI module and TJA1051 CAN IC, supporting up to 30V input voltage.
- Developed a Python tool for synchronized pose and camera data capture on a NVIDIA Jetson based ROS2 platform.
- Created and tested a 1D Bidirectional CNN model for hand gesture recognition using pose data.
- Applied image processing techniques to detect cutouts on display screens to detect bugs and verify manufacture quality.

Monterey Bay Aquarium Research Institute

Moss Landing, California

Computer Vision Intern

June 2023 - August 2023

- Developed OpenCV based realtime disparity and distance estimation tool for fisheye stereo cameras with accuracy range of 5 meters (+- 300 mm) for desktop and VR.
- Used multiprocessing to perform disparity and tracking in parallel.
- Integrated FathomNet YOLOv5 deep sea organism tracking model with distance estimation.
- Used Sockets to send realtime frames, and distance information between Python OpenCV application and Unity VR.